Leps-P: Probabilistic Strength Envelopes with Stress-Decaying Curvature for Rockfill and Mine Waste Rock

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Abstract

Reliability-based design of high waste dumps needs the drained shear strength of rockfill and mine waste rock as a distribution, with its dependence on stress and a rule to combine published experience with site tests. Leps-P is a three-level hierarchical model of the secant friction angle in which the dilatant part of strength decays with stress toward an asymptote; each material has its own dilatant level and decay rate. It was calibrated on 767 large-scale triaxial tests from 36 publications of the public Leps 2.0 database. The likelihood favored this form over a quadratic log–log envelope with the same number of parameters (log-likelihood difference 9.8) and over a form without an asymptote (44.7). The asymptote, about 25°, is a population decay parameter well below the angles measured at the highest stresses, not a critical-state angle. Leaving out one publication at a time, the bounded, quadratic, and Leps 2.0 envelopes predicted the strength of new materials with statistically indistinguishable accuracy, but only Leps-P had calibrated intervals at low stress. With a few site tests at low stress, updating the level of each material gave calibrated predictions at higher stress (83% coverage of the 80% interval); differences from simpler fits were not significant with 13 publications. At field scale, about 62% of the variance of the strength of an untested waste rock at 3 MPa came from the assumed heterogeneity of the specimen-size effect. For a 700 m dump on a competent foundation, analyzed with the Morgenstern–Price method, the factor of safety was 1.47 with the median and 1.20 with the 5th-percentile envelope of placed waste rock. Design charts and open code are provided.

Practical Applications

Designers of high waste dumps and rockfill dams often have few or no large-scale shear tests and rely on published charts of rockfill strength. This work turns the Leps chart into a tool that gives a best estimate of strength and a cautious value for design, for any rock quality and density, together with the range of uncertainty. Its curves flatten at high pressure, as expected when crushed rock stops dilating, and they are supported by tests up to about 8 MPa on the failure plane. A few tests of the site material at low pressure sharpen the estimate at higher pressure, and the tool warns when site tests disagree with published experience. For untested waste rock, most of the remaining uncertainty comes, under the adopted model, from the size of the rock particles and from differences between materials and laboratories; testing the actual waste rock at large scale and at two particle sizes is therefore the most valuable investigation. Free software accompanies the paper.

Author keywords: Rockfill; Mine waste rock; Shear strength; Nonlinear failure envelope; Dilatancy; Hierarchical model; Reliability-based design; High waste dumps.

Introduction

Leps (1970) compiled 109 large-scale triaxial tests and proposed three nonlinear envelopes for rockfill that have been used for dams and waste dumps whenever site-specific tests were unavailable. Cylwik et al. (2025) updated them with a public database of 938 tests (Arrieta 2025), adopted the power criterion of De Mello (1977) and Charles and Watts (1980) with a single fitting parameter, the secant friction angle at $\sigma_n' = 10p_a$, and related that angle to rock quality and void ratio. That update is referred to here as Leps 2.0.

Two developments motivate a further extension. First, design is moving toward reliability-based criteria (Read and Stacey 2009; Hawley and Cunning 2017), which require a

strength distribution and a rule to combine published experience with site tests. Second, waste dumps now reach 500–700 m (Valenzuela et al. 2008). For a uniform dump on a competent foundation the critical surface stays at normal stresses of a few megapascals, but mechanisms controlled by the foundation, by a basal layer, or by the weakest material reach higher stresses, where waste rock crushes and loses dilatancy (Marsal 1973; Hardin 1985; Lade et al. 1996; Linero et al. 2007; Ovalle et al. 2020). How an envelope behaves at the edge of and beyond the data then matters: a power law with a constant exponent keeps curving downward, whereas the excess of strength over the critical state, which is the contribution of dilatancy, is suppressed by confinement (Bolton 1986; Jefferies and Been 2016).

Hierarchical models that borrow strength from generic databases have been developed for soil properties (Ching and Phoon 2019; Ching et al. 2021), and a hierarchical Bayesian analysis of the Barton and Kjærnsli (1981) parameters has been applied to the rockfill groups of a single project (Contreras et al. 2022; Contreras and Linero-Molina 2025). Leps-P differs in pooling 36 publications, in its envelope form, and in its out-of-sample evaluation. Its contributions are (1) an envelope whose curvature decays with stress for every material, compared with fixed-form alternatives by likelihood and out of sample; (2) the separation of publication and material variability; (3) a site-updating rule with a prior–data conflict check, compared with the simple fits used in practice; (4) a decomposition of the uncertainty of an untested waste rock at field scale; and (5) design charts and an open implementation.

Database and Model Formulation

Calibration Database

The public database (Arrieta 2025) was filtered to tests with $\sigma_3' \geq 60$ kPa, reported initial void ratio, and at least two tests per set, giving 767 tests on 226 sets from 36 publications (Fig.

S1). A set is a material with a given gradation, density, and rock type tested in one program; failure is the maximum deviator stress, as in Leps 2.0. Rock quality classes follow Table 5 of Cylwik et al. (2025). Tests with σ_n' > 3 MPa (90 tests) come from eight publications, seven of them published between 1963 and 1983 (e.g., Marachi et al. 1972; Marsal 1973), and sets with ϕ_1' < 31° were excluded by Cylwik et al. (2025), so the lower tail of the population is truncated. At the maximum deviator the specimens were contracting: the median net volumetric strain was -0.9% below 0.5 MPa and -6.9% above 3 MPa, where 99% of the specimens contracted and the median breakage index of Marsal (1973) was 36% (Fig. S2). Above 5 MPa the measured secant angles ranged from 35.5° to 41.2° (5th to 95th percentile, 34 tests).

Envelope

The Leps 2.0 criterion is the power law

$$\tau = \tan\phi_1' \sigma_0^{1-m} {\sigma_n'}^m \quad (1)$$

where $\sigma_0 = 10p_a$, p_a = 0.1013 MPa, ϕ_1' is the secant friction angle at $\sigma_n' = \sigma_0$, and m is a constant exponent; the secant friction angle is $\phi_s' = \arctan(\tau/\sigma_n')$. In granular materials the excess of strength over the critical state is governed by dilatancy, which is controlled by the state of the material relative to its critical state and is suppressed by confinement (Bolton 1986; Jefferies and Been 2016, p. 43); particle crushing reinforces this trend (Hardin 1985; McDowell and Bolton 1998; Coop et al. 2004; Einav 2007). A minimal form with this property is

$$\tan\phi_s' = T + (\tan\phi_1' - T)\exp(-\zeta x), x = \ln(\sigma_n'/\sigma_0), T = \tan\phi_\infty' \quad (2)$$

in which the dilatant part decays at rate ζ toward the asymptotic angle ϕ_∞'. For ϕ_∞' = 0 it reduces to a power law with $m = 1 - \zeta$; Bolton (1986) proposed instead a dilatant part that vanishes at a finite stress. Because the specimens at high stress were contracting at the maximum deviator,

part of the curvature reflects strain-limited, under-mobilized tests, so ϕ'_∞ is a population decay parameter of the maximum-deviator envelope and not a critical-state angle.

Hierarchical Model

With $y = \ln(\tau/\sigma_0)$, the strength of test i of set j of publication k is

$$y_{ijk} = x_{ijk} + \ln\left[T + \left(\exp\eta_{jk} - T\right)\exp\left(\quad -\zeta_{jk} x_{ijk}\right)\right] + \varepsilon_{ijk} \tag{3}$$

$$\eta_{jk} = \beta_{c(jk)} + \beta_e e_{jk} + u_{\eta,jk}, \zeta_{jk} = \zeta_0 + u_{\zeta,jk}, \mathbf{u}_{jk} = \mathbf{u}_k^{(p)} + \mathbf{u}_{jk}^{(s)} \tag{4}$$

$$\mathbf{u}^{(p)} \sim N(\mathbf{0}, \mathbf{G}_p), \mathbf{u}^{(s)} \sim N(\mathbf{0}, \mathbf{G}_s), \ln\sigma_\varepsilon(x) = a_0 + a_1 x \tag{5}$$

where $\eta = \ln \tan \phi'_1$ is the dilatant level of the material, β_c the coefficient of rock quality class c, β_e the effect of initial void ratio e, ζ_0 the population decay rate, $\mathrm{u} = (u_\eta, u_\zeta)$ the publication and set random effects with covariance matrices $\mathbf{G}_p$ and $\mathbf{G}_s$, and ε the test-to-test deviation with standard deviation σ_ε (Laird and Ware 1982). Every material thus has its own envelope that decays toward the common asymptote, and its local slope is below one when $\zeta > 0$. The publication effect combines laboratory, procedure, era, and material selection; because true strength and laboratory bias cannot be distinguished, it is treated as uncertainty of strength. Through β_c and β_e the median ϕ'_1 reproduces the relation $\phi'_1 = \phi'_{e=0} - \omega e$ of Leps 2.0. A quadratic log–log form, $y = \theta + m \cdot x + c_2 x^2$, and a power form ($c_2 = 0$), with random effects on θ and m, were fitted for comparison.

The asymptote is weakly identified: lower values of ϕ'_∞ with slower decay fit nearly as well as higher values with faster decay. The profile likelihood of (ϕ'_∞, ζ_0) was evaluated on a grid (14–32° and 0.175–0.475), the points holding 99.5% of the normalized likelihood were refitted, and predictions average over them with weights proportional to the likelihood (flat prior on the grid); a prior uniform in $\tan \phi'_\infty$ was used as a sensitivity.

Predictive Distribution and Design Values

For a new material of class c and void ratio e tested by a new laboratory, y is approximately normal with mean $f(x)$ given by Eq. (3) with u = 0 and variance

$$\mathrm{Var}[y] = \mathbf{z}^T(\mathbf{G}_p + \mathbf{G}_s)\mathbf{z} + \mathbf{j}^T\mathbf{C}_\beta\mathbf{j} + \sigma_\varepsilon^2(x) \tag{6}$$

where $\mathbf{z}$ and $\mathbf{j}$ are the gradients of f with respect to (η, ζ) and to the fixed effects and $\mathbf{C}_\beta$ is the covariance of the fixed effects (first-order linearization; Lindstrom and Bates 1990). The first two terms describe the uncertainty of the mean envelope of the material, which does not average out along a failure surface within that material (Christian et al. 1994); the last describes test-to-test scatter. The implementation samples the variance parameters from their posterior, (ϕ'_∞, ζ_0) from the ridge weights, the fixed effects, and the random effects. Fig. 1 and Table 3 give the median and 5th percentile of the mean envelope of untested materials at laboratory specimen size.

Updating with Site Tests

Site triaxial tests are reduced with the maximum-deviator criterion of the database and converted to (σ'_n, τ) with Eqs. (13), (14), and (16) of Cylwik et al. (2025). Consistency with the prior is checked with the prior predictive Mahalanobis distance

$$D^2 = \mathbf{r}^T\mathbf{V}^{-1}\mathbf{r} \sim \chi^2_n, \mathbf{V} = \mathbf{Z}(\mathbf{G}_p + \mathbf{G}_s)\mathbf{Z}^T + \mathbf{J}\mathbf{C}_\beta\mathbf{J}^T + \mathbf{S} \tag{7}$$

where $\mathbf{r}$ holds the residuals r_i of the n site tests from the prior mean, $\mathbf{Z}$ and $\mathbf{J}$ have rows $\mathbf{z}_i^T$ and $\mathbf{j}_i^T$, and $\mathbf{S} = \mathrm{diag}\sigma_\varepsilon^2(x_i)$; a p-value below 0.05 flags the tests for review. The update treats the residuals as measurements of the level of the envelope at the centroid $\overline{x}$ of the tests. With $\mathbf{v}$ the random effects of the material and the deviation of the fixed effects, $\mathbf{P}$ their prior covariance, $\mathbf{a}$ the gradient at $\overline{x}$, and $g_0 = \mathbf{a}^T\mathbf{P}\mathbf{a}$, the level $q_0 = \mathbf{a}^T\mathbf{v}$ has the posterior

$$\hat{q}_0 = \frac{\sum_i r_i/\sigma_{\varepsilon,i}^2}{1/g_0+\sum_i 1/\sigma_{\varepsilon,i}^2}, \mathrm{Var}(q_0) = \left(\frac{1}{g_0} + \sum_i \frac{1}{\sigma_{\varepsilon,i}^2}\right)^{-1} \quad (8)$$

and $\mathbf{v}$ is shifted by $\mathbf{P}\mathbf{a}\hat{q}_0/g_0$. Eq. (8) is a cut (modular) update that can be applied in a spreadsheet; it does not estimate the decay of a single material from a few tests. When several materials of one site were tested by the same laboratory, flagged sets are excluded and the remaining sets are updated jointly with a shared publication effect, which flagged and untested materials then inherit.

Placement Density and Specimen Size

A change of void ratio Δe from the specimens to the placed material enters through η, so its effect fades with stress. Because β_e was estimated between sets, in which void ratio also reflects gradation and particle shape, applying it to a change of state of one material is an assumption. Specimen size enters as a shift of level

$$\Delta y = \beta_D \ln\left(D_f/D_l\right) \quad (9)$$

where D_f and D_l are the maximum particle sizes in the field and in the laboratory; Weibull scaling predicts $\beta_D < 0$ (Frossard et al. 2012; Ovalle et al. 2014; Alonso et al. 2012).

Estimation and Out-of-Sample Evaluation

Fixed effects were estimated by Gauss–Newton iteration on the linearized mean, with restricted maximum likelihood (REML) for the variance parameters. Variance parameters were sampled with an adaptive Metropolis algorithm under weakly informative priors (half-normal with scale 0.25 on standard deviations, uniform on correlations, normal on a_0, a_1), with four chains from dispersed starting values; the fixed effects are thus estimated conditionally (an empirical Bayes scheme). Convergence was checked with the rank-normalized potential scale reduction factor and bulk and tail effective sample sizes (Vehtari et al. 2021).

Each publication was left out in turn; in each fold (ϕ'_∞, ζ_0) were selected on the training data only and the model was refitted. In Mode A (no site tests), every test of the held-out publication was predicted. In Mode B, each held-out set with at least two tests at $\sigma'_n \leq 1.5$ MPa and one at $\sigma'_n \geq 2$ MPa was updated with the former, and the latter were predicted with: the prior; the level update of each set (Eq. 8); the same after discarding flagged sets; the recommended procedure (flagged sets discarded and all sets of the publication updated jointly with a shared laboratory effect); a joint update of both random effects; and least-squares fits of ϕ'_1 with m = 0.90 or with m equal to the median within-set slope of the training data, whose predictive standard deviation was their error on the training publications. These procedures were developed while examining earlier versions of the model on the same held-out sets, so the Mode B comparisons are exploratory. Scores were the coverage of the 80% interval, the fraction of tests below the 10th percentile, the continuous ranked probability score (CRPS) of ln τ (Gneiting and Raftery 2007), and, for the 10th percentile, the quantile (pinball) loss. Differences in publication means were tested with exact sign-flip tests, with Holm (1979) adjustment within each mode, and summarized with bootstrap intervals over publications. The Mode A reference was the Leps 2.0 envelope (m = 0.9) with a constant dispersion fitted to the training data; its constants were fitted to the whole database, which favors it.

The folds used plug-in predictive distributions. With all data, the distribution of the implementation differed from the plug-in one by at most 0.005 in median ln τ and had a standard deviation 1.03–1.09 times as large; the implementation uses the same update rule as Mode B.

Results

Parameters and Evidence for Stress-Decaying Curvature

Table 1 lists the parameters. The rank-normalized potential scale reduction factor was at most 1.017, and the bulk and tail effective sample sizes were at least 502 and 849. Half-normal prior scales of 0.1 and 0.5 changed the median between-publication standard deviation of η from 0.061 to 0.061 and 0.062. At σ_0, publication effects account for 35% (90% credible interval 19%–51%) of the between-material variance of η.

The bounded form exceeded the quadratic form, which has the same number of parameters, by 9.8 in maximum log-likelihood, and removing the asymptote ($\phi'_\infty = 0$, a power law in tan ϕ'_s) reduced it by 44.7. Within 3 log-likelihood units of the maximum, ϕ'_∞ ranged from 20° to 27° (Fig. 2); the weighted mean was 24.6° (24.6° with a prior uniform in tan ϕ'_∞) and $\zeta_0 = 0.30$. All 36 folds selected ϕ'_∞ of 24–26° and $\zeta_0 = 0.30$. These are in-sample comparisons; out of sample the forms were indistinguishable (next section).

Mode A: No Site Tests

Across all held-out tests (Table 2, Fig. 3a), the bounded and quadratic forms had CRPS 0.0680 and 0.0688 (−0.0008 (95% CI −0.0036 to +0.0015; sign-flip $p = 0.56$, Holm-adjusted 1.00)), and Leps-P covered 84% of tests with its 80% interval. Above 2.75 MPa (100 tests, 9 publications) both forms underpredicted the median (mean standardized residual +0.21 and +0.28), and the Leps 2.0 reference had a lower 10% quantile loss (+0.0033 (95% CI +0.0014 to +0.0056; sign-flip $p = 0.01$, Holm-adjusted 0.14)). Below 0.5 MPa the constant-dispersion interval covered only 70% of tests, with the median biased low (mean standardized residual +0.59), whereas Leps-P covered 84%; this reflects the constant dispersion of the reference, not the Leps 2.0 envelope itself.

Mode B: Site Tests at Low Stress

With the level update of each set, 83% of the held-out tests fell within the 80% interval and the standard deviation of standardized residuals was 0.91, so the updated intervals were calibrated (Table 2, Fig. 3b). The CRPS was 0.0308, against 0.0424 for the prior, 0.0369 for the least-squares fit with m from the training data, and 0.0407 for the fit with $m = 0.90$, which placed 17% of tests below its 10th percentile. The recommended procedure scored 0.0344 and the joint update 0.0330. None of these differences was significant after adjustment (smallest Holm-adjusted $p = 0.48$); with 64 tests from 13 publications the comparison can detect only large differences. The conflict check flagged 9% of the sets.

Specimen Size

Sixteen materials of 12 publications were tested at maximum particle sizes differing by a factor of two or more. With set intercepts adjusted for void ratio and materials combined within publications, the random-effects mean of β_D was +0.041 (Knapp–Hartung 95% CI −0.001 to +0.084; Knapp and Hartung 2003) with a between-publication standard deviation $\tau = 0.063$ (Q-profile 95% CI 0.042–0.107; Viechtbauer 2007; Fig. S4). Adding the between-set variance of the model to each estimate gave $\tau = 0.049$. The interval for the mean excludes the magnitude of about −0.05 to −0.15 implied by Weibull scaling of homothetic gradations, while the database mixes scalped and parallel gradations and rounded and angular particles, for which opposite trends have been reported (Varadarajan et al. 2003). For design, Leps-P uses a zero mean with the larger τ as uncertainty.

Sources of Uncertainty at Field Scale

For an untested material of low rock quality placed at $e = 0.65$ and scaled from 0.18 m to 1 m particles, the standard deviation of the mean-envelope ln τ at 3 MPa corresponds to about

3.8° in ϕ_s'. Under the adopted size model, specimen size contributed 62% of the variance, between-set variability 24%, publication 12%, the asymptote and decay rate 0%, and the fixed effects 2% (Fig. S3); at 10 MPa the asymptote contributed 2%. The size share scales with $\ln^2(D_f/D_l)$. Because all materials share the asymptote, between-material variance decreases at very high stress; this is a property of the model, not an observation.

Application: Waste Rock from a Large Open-Pit Mine

Site Data and Conflict Check

The waste rock of a large open-pit mine comprises five rock types, A to E. Drained triaxial tests on large specimens of scalped gradation by one laboratory were available for A, B, and C (5 sets); D and E were untested, and E is prone to degradation. Rock quality was taken as low for A, C, D, and E and moderate for B; illustrative mass fractions of 35%, 25%, 20%, 10%, and 10% were used. For rock type A the conflict check gave $p = 0.18$ with the low class and 0.026 with the good class, so the tests were consistent with the class assigned. The check flagged 3 of the 5 sets (Fig. 4a): both sets of B ($p \leq 0.0001$) and one of C ($p = 0.025$). In the B sets the secant angle increased with stress and the specimens were still contracting at the maximum deviator, reached at 13–19% axial strain. Re-reducing them from the tabulated stress–strain curves with the deviator at 10% and 15% axial strain gave free log–log slopes of 1.04–1.19, each from three tests, so the increase is not an artifact of the failure definition, although slopes near one are not significantly above it. A slope above one is excluded at the population level by the model, not for individual materials. Fixed-strain criteria created new conflicts in the other rock types, which confirms that site tests must be reduced with the criterion of the calibration database. Flagged sets did not update the envelope, and the remaining sets were updated jointly with a common laboratory effect; each rock type is represented by one material.

Placement Density, Specimen Size, and Dump-Average Envelope

Laboratory specimens had void ratios of 0.31–0.42; a placement void ratio of 0.65 was adopted for end-dumped waste rock, with an additional increase of 0.15 for the degradation of E. A failure surface through a mixed dump mobilizes the mass-weighted strength of the rock types it crosses; this assumes mixing along the surface, and the weakest-rock-type case bounds zoned or segregated placement. At σ_n' = 3 MPa the median and 5th percentile of the dump-average ϕ_s' were 37.4° and 35.9° at laboratory density and size, 33.4° and 32.0° at placement density without size shift, and 33.4° and 28.4° with both (Table 4; Fig. 4b). Density thus lowered the median and size uncertainty the 5th percentile. A surface confined to E lowered the 5th percentile by 2.7°, and giving D and E a laboratory effect independent of the site changed it by 0.3°.

Stability of a High Dump

Dumps 150–700 m high with an overall slope of 2H:1V on a competent, drained, level foundation were analyzed in Slide2 (Rocscience year) with the Morgenstern and Price (1965) method (half-sine interslice function) and a noncircular search optimized by simulated annealing, using the tabulated envelopes (Table 5). The unit weight was consistent with the void ratio: 17.2 kN/m³ for e = 0.65 (specific gravity 2.75, water content 5%), 15.7 kN/m³ for E (e = 0.80), and 21 kN/m³ at laboratory density. For 700 m, the factor of safety was 1.47 with the median and 1.20 with the 5th-percentile dump-average envelope; at placement density without the size shift it was 1.47 and 1.38, and at laboratory density and size 1.69 and 1.60. The quadratic form gave 1.46 and 1.20: on a competent foundation the critical surface stays at a few megapascals, where the forms agree. In a Monte Carlo analysis, each of 600 envelope samples was analyzed in Slide2 with its own search of the critical surface. The model correlation gave 1 failure (probability of failure 0.0–0.9%, Clopper–Pearson 95% interval; Clopper and Pearson 1934; lognormal

reliability index 3.3), and a surface confined to E gave 15 of 600 (1.4–4.1%; index 2.0). These are conditional, strength-only probabilities for a drained foundation and a two-dimensional section, which exclude placement density, unit weight, and foundation as random variables.

Discussion

What Leps-P Adds

Leps-P does not predict the central strength of an untested material better than Leps 2.0, and with few site tests its advantage over simple fits is not statistically demonstrable with the available publications. Its contributions are a material-level envelope whose curvature decays with stress, strongly favored by the likelihood and consistent with the suppression of dilatancy; intervals that are calibrated where a constant dispersion is not, both without and with site tests; an updating rule with a conflict check; and a decomposition showing that, under the adopted size model, particle size and the identity of the material dominate the uncertainty of an untested waste rock. Large-scale tests of the actual waste rock, and of one material at two particle sizes, are therefore the most valuable investigations.

Design Format

A procedure consistent with these results is: (1) assign rock quality and placement void ratio to each rock type, reduce any site tests with the maximum-deviator criterion, and check the conflict diagnostic; (2) update with the tests that pass (Eq. 8); (3) shift the envelopes to placement density and field particle size; (4) adopt as characteristic value for a global mechanism the 5th percentile of the mean envelope, a Bayesian cautious estimate of the mean (Prästings et al. 2019) in the spirit of Eurocode 7 (CEN 2004, 2024); and (5) check the probability of failure by sampling the envelopes with stated correlations between rock types, including the weakest-rock-type case. Because the factor of safety in limit equilibrium divides the shear strength, the

factor computed with the characteristic envelope equals the strength factor that the design would still tolerate; applying mining acceptance criteria calibrated for best-estimate strength (Read and Stacey 2009; Hawley and Cunning 2017) on top of it would partly count the strength uncertainty twice, since those criteria also cover model, pore-pressure, and geometric uncertainty. No target values are proposed here; the pairing of factor of safety and conditional probability of failure in Table 5 is an example to be calibrated site by site.

Limitations

The validity domain is $\sigma_n' \leq 8$ MPa and, above 3 MPa, $e \leq 0.55$, where 83 of the 90 high-stress tests lie; percentiles beyond it are model extrapolations. Tests above 3 MPa come from eight publications, the lower tail of the population is truncated at $\phi_1' = 31°$, class labels may carry information from measured strength, and the within-material residual is laboratory scatter rather than spatial variability. The size-effect heterogeneity, the largest term at field scale, rests on 12 publications. The site application uses one laboratory and illustrative proportions. Pore pressure in saturated basal layers, wetting collapse (Oldecop and Alonso 2001), seismic loading, three-dimensional geometry, and static liquefaction are outside the scope. Leps-P can be re-estimated as drained tests at σ_3' of 5–10 MPa, tests of one material at two sizes, and in situ density data become available.

Conclusions

1. An envelope in which the dilatant part of tan ϕ_s' decays with stress toward an asymptote, with a material-level dilatant level and decay rate, was strongly favored by the likelihood over fixed-form alternatives; the asymptote (about 25°) is a decay parameter, not a critical-state angle.

2. Out of sample, the bounded, quadratic, and Leps 2.0 envelopes predicted new materials with indistinguishable accuracy; Leps-P intervals were calibrated at low stress, where a constant dispersion was not.
3. Updating the level of each material with a few site tests at low stress gave calibrated predictions at higher stress; its advantage over simpler fits was not statistically significant with 13 publications.
4. No systematic specimen-size effect was detected within materials; under the adopted model its heterogeneity dominates the uncertainty of an untested waste rock at field scale.
5. For a 700 m dump on a competent foundation, analyzed with the Morgenstern–Price method, placement density lowered the median strength and size uncertainty the cautious strength; the envelope form did not affect the factor of safety.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available in a repository or online in accordance with funder data retention policies (Leps-P code, parameters, and evaluation scripts: repository DOI; calibration database: Arrieta 2025). Some or all data, models, or code generated or used during the study are proprietary or confidential in nature and may only be provided with restrictions (the site test results, which may be provided in anonymized form with the permission of their owner).

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Notation

The following symbols are used in this paper:

$\mathbf{C}_\beta$ = covariance matrix of the fixed effects;

D_f, D_l = maximum particle size in the field and in the laboratory;

D^2 = prior predictive Mahalanobis distance of the site tests;

$\mathbf{G}_p$, $\mathbf{G}_s$ = covariance matrices of the publication and set random effects;

J, **Z** = matrices of the gradients of the envelope with respect to the fixed and random effects;

P = prior covariance of the random effects and fixed-effect deviations of a material;

S = diagonal matrix of test-to-test variances;

T = tan ϕ'_∞;

V = prior predictive covariance of the site tests;

a = gradient of the envelope at the centroid of the site tests;

a_0, a_1 = parameters of the test-to-test dispersion;

c, c_2 = rock quality class; quadratic coefficient of the comparison form;

e = initial void ratio;

f = mean envelope of a new material in terms of y;

g_0 = prior variance of q_0;

j, **z** = gradients of f with respect to the fixed and random effects;

m = exponent of the power envelope;

n = number of site tests;

p_a = atmospheric pressure (0.1013 MPa);

q_0 = level of the envelope at the centroid of the site tests;

r, r_i = residuals of the site tests from the prior mean;

$\mathbf{u}$, $\mathbf{u}^{(p)}$, $\mathbf{u}^{(s)}$ = random effects on (η, ζ): total, publication, and set;

$\mathbf{v}$ = random effects and fixed-effect deviations of a material;

$x = \ln(\sigma_n'/\sigma_0)$;

$\overline{x}$ = mean of x over the site tests of a set;

$y = \ln(\tau/\sigma_0)$;

β_c = coefficient of rock quality class c;

β_D = specimen-size coefficient;

β_e = effect of void ratio on η;

Δe, Δy = change of void ratio; shift of y for specimen size;

ε = test-to-test deviation;

ζ, ζ_0 = decay rate of a material and of the population;

η = dilatant level of a material, ln tan ϕ_1';

θ = level of the comparison forms;

σ_0 = reference stress, $10p_a$;

σ_3' = effective confining stress;

σ_n' = effective normal stress on the failure plane;

σ_ε = standard deviation of test-to-test scatter;

τ = shear strength; in the size analysis, between-publication standard deviation of β_D;

ϕ_1' = secant friction angle at $\sigma_n' = \sigma_0$;

ϕ_∞' = asymptotic secant friction angle;

$\phi_{e=0}' = \phi_1'$ extrapolated to $e = 0$ in Leps 2.0;

ϕ_s' = secant friction angle;

ω = decrease of ϕ_1' per unit void ratio in Leps 2.0;

Supplemental Materials

Figs. S1–S4 (stress coverage of the database; volumetric strain and breakage at failure; variance decomposition; specimen-size effect by material) and Table S1 (every Mode B procedure evaluated) are available online in the ASCE Library (www.ascelibrary.org).

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Figure Captions List

Fig. 1. Design envelopes of Leps-P for a new material tested by a new laboratory, at laboratory specimen size, by rock quality and void ratio: median (thick lines) and 5th percentile (thin lines) of the mean envelope. The shaded zone exceeds the largest normal stress in the database.

Fig. 2. Profile log-likelihood of the asymptotic angle and the population decay rate relative to its maximum (dot).

Fig. 3. Out-of-sample evaluation, leaving out each publication: (a) coverage of the 80% interval by range of normal stress without site tests; and (b) mean CRPS of ln τ for held-out tests at $\sigma_n' \geq$ 2 MPa with site tests at $\sigma_n' \leq 1.5$ MPa (bars: 95% bootstrap intervals over publications; filled: recommended procedure).

Fig. 4. Waste rock of the example: (a) laboratory-scale envelopes by rock type with the site tests (open symbols: tests flagged by the conflict check); and (b) envelopes of placed waste rock (void ratio 0.65, maximum particle size 1 m) by rock type and dump-average envelope. Dotted vertical lines mark the largest normal stress of the site tests.

Table 1. Parameters of Leps-P

Parameter	Estimate	Standard error or 90% credible interval
β_c, good rock quality	0.231	0.034
β_c, moderate rock quality	0.098	0.028
β_c, low rock quality	0.023	0.024
β_e	−0.545	0.050
ϕ'_∞ (°), weighted mean over the ridge	24.6	20–27 (within 3 log-likelihood units)
ζ_0, weighted mean over the ridge	0.30	—
SD of η, publication	0.061	0.042–0.083
SD of ζ, publication	0.073	0.050–0.105
Correlation of η and ζ, publication	0.17	−0.27 to 0.55
SD of η, set	0.083	0.075–0.091
SD of ζ, set	0.068	0.055–0.082
Correlation of η and ζ, set	−0.11	−0.29 to 0.09
σ_ε at $\sigma'_n = \sigma_0$	0.035	0.033–0.038
a1	−0.234	−0.305 to −0.165

Note: 767 tests, 226 sets, 36 publications. Fixed effects at the maximum-likelihood point of the ridge (ϕ'_∞ = 25°, ζ_0 = 0.3); variance parameters: posterior median and 90% credible interval. Median ϕ'_1 at e = 0: 51.6°, 47.8°, and 45.7° for good, moderate, and low rock quality.

Table 2. Out-of-sample evaluation (leave one publication out)

Procedure	80% coverage	Below P10	CRPS
Mode A, all stresses			
Leps-P, bounded form	84%	10%	0.0680
Leps-P, quadratic form	83%	9%	0.0688
Leps 2.0 envelope, constant dispersion	82%	6%	0.0696
Mode A, σ'_n > 2.75 MPa			
Leps-P, bounded form	88%	4%	0.0478
Leps-P, quadratic form	88%	4%	0.0468
Leps 2.0 envelope, constant dispersion	96%	4%	0.0446
Mode B, site tests at $\sigma'_n \leq$ 1.5 MPa			
Prior, no site tests	95%	3%	0.0424
Fit of ϕ'_1, m = 0.90	81%	17%	0.0407
Fit of ϕ'_1, m from training data	81%	2%	0.0369
Joint update of both random effects	80%	12%	0.0330
Level update of each set	83%	9%	0.0308
Recommended	83%	12%	0.0344

procedure

Note: Mode A: 767 tests (36 publications); above 2.75 MPa, 100 tests (9 publications). Mode B: 64 tests at $\sigma'_n \geq 2$ MPa (46 sets, 13 publications). Nominal values: 80% coverage and 10% below P10. CRPS in units of ln τ. No difference was significant after Holm adjustment.

Table 3. Design values of the secant friction angle ϕ'_s (°) for an untested material: median / 5th percentile of the mean envelope

Rock quality	e	0.5 MPa	1 MPa	3 MPa	5 MPa	8 MPa
Good	0.30	50.4 / 44.6	46.9 / 41.9	42.0 / 37.3	40.0 / 35.3	38.3 / 33.6[a]
	0.45	47.9 / 42.1	44.6 / 39.6	40.0 / 35.6	38.2 / 33.8	36.6 / 32.3[a]
	0.65	44.5 / 38.7	41.5 / 36.6	37.4 / 33.3	35.8 / 31.9[a]	34.5 / 30.7[a]
Moderate	0.30	46.3 / 40.5	43.2 / 38.3	38.8 / 34.5	37.1 / 32.9	35.6 / 31.6[a]
	0.45	43.7 / 38.0	40.8 / 36.0	36.9 / 32.9	35.4 / 31.5	34.1 / 30.4[a]
	0.65	40.2 / 34.6	37.8 / 33.1	34.5 / 30.7	33.2 / 29.7[a]	32.2 / 28.8[a]
Low	0.30	43.9 / 38.2	41.0 / 36.2	37.1 / 33.0	35.5 / 31.6	34.2 / 30.5[a]
	0.45	41.3 / 35.7	38.7 / 34.0	35.2 / 31.4	33.9 / 30.3	32.7 / 29.3[a]
	0.65	37.9 / 32.4	35.7 / 31.1	32.9 / 29.3	31.8 / 28.5[a]	30.9 / 27.8[a]

Note: Laboratory specimen size, new laboratory, no site tests, drained, maximum-deviator criterion; for global mechanisms. Field values are lower when the specimen-size uncertainty is added (about 3° at the 5th percentile for a size ratio of 5.6). Not applicable to degradable, argillaceous, or fines-rich waste below the database range ($\phi'_1 \geq 31°$). [a]Extrapolated beyond the validity domain ($\sigma'_n \leq 8$ MPa; e ≤ 0.55 above 3 MPa).

Table 4. Secant friction angle ϕ'_s (°) of the waste rock of the example, median / 5th percentile of the mean envelope

Material	Laboratory, 3 MPa	Placed, 1 MPa	Placed, 3 MPa
Rock type A	36.5 / 35.5	35.3 / 30.2	32.5 / 27.8
Rock type B	38.7 / 34.7	38.2 / 31.5	34.8 / 28.8
Rock type C	37.7 / 36.5	37.2 / 32.0	34.1 / 29.2
Rock type D (untested)	36.9 / 33.2	36.1 / 29.4	33.2 / 27.2
Rock type E (untested)	36.9 / 33.2	33.8 / 27.4	31.5 / 25.7
Dump average, model correlation	37.4 / 35.9	36.4 / 31.1	33.4 / 28.4
Dump average, full	37.4 / 35.0	36.4 / 30.6	33.4 / 28.1

correlation

Note: Laboratory: specimen void ratio and size. Placed: void ratio 0.65 (rock type E, 0.80), maximum particle size 1 m, size effect with zero mean. The full-correlation case sorts the samples of each rock type at every stress, an upper bound on dependence. No site tests above σ_n' ≈ 4.6 MPa.

Table 5. Factor of safety of dumps on a competent level foundation (Morgenstern–Price, Slide2), placed waste rock: median / 5th-percentile envelope

Envelope	150 m	400 m	700 m
Dump average, model correlation	1.76 / 1.41	1.56 / 1.27	1.47 / 1.20
Dump average, full correlation	1.75 / 1.36	1.56 / 1.25	1.47 / 1.19
Weakest tested rock type only	1.68 / 1.34	1.50 / 1.23	1.42 / 1.17
Untested, degraded rock type E only	1.58 / 1.18	1.43 / 1.10	1.36 / 1.06

Note: Unit weight 17.2 kN/m³ (rock type E only: 15.7 kN/m³). Monte Carlo at 700 m with 600 envelope samples, each with its own search: model correlation 1 failure (0.0–0.9%; lognormal reliability index 3.3); rock type E only 15 failures (1.4–4.1%; index 2.0). 700 m, model correlation: without size shift 1.47 / 1.38; laboratory density and size (21 kN/m³) 1.69 / 1.60; quadratic form 1.46 / 1.20.